

Importance of Multidisciplinary Design Optimization for Wave-Driven Desalination Systems

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Abstract

This is the abstract. Summarize your problem, approach, and key results concisely. You can include equations and references if needed.

1. Introduction

The global freshwater crisis necessitates innovative solutions for clean water production. Demand for freshwater is anticipated to grow by over 40% by 2050 [1], while droughts, urbanization, and uneven distribution of water resources add to the pressure on our existing freshwater supplies. Seawater reverse osmosis (SWRO) is a proven technology for creating freshwater from seawater, but is quite energy intensive, requiring 2-4 kWh/m³ [2]. This cost associated with this high energy demand make up 25%-40% of the total cost of water [3], and when supplied through fossil fuels, this energy consumption further exacerbates water scarcity challenges [4]. To ensure sustainability, it is therefore crucial to power SWRO efficiently through renewable energy sources.

Wave energy converters (WECs) are an intriguing solution. SWRO and WECs have natural co-location, with SWRO plants typically situated near the coast while WECs are deployed offshore. Beyond both being marine-based technologies, WECs and SWRO plants share complementary operational characteristics [3]. SWRO requires high pressure seawater, and WECs can provide this directly using a hydraulic style power take-off (PTO) system, forming a wave driven desalination system [5]. This direct drive approach eliminates the need for conversion to an intermediate form of energy, potentially improving overall system efficiency and reducing system complexity. This has the potential to give WECs a significant advantage over other renewable energy sources for powering SWRO desalination.

Despite their potential, WECs face significant challenges limiting their widespread adoption. Notably, their high costs pose a major barrier. Therefore, design optimization is essential to improve economic viability. Multidis-

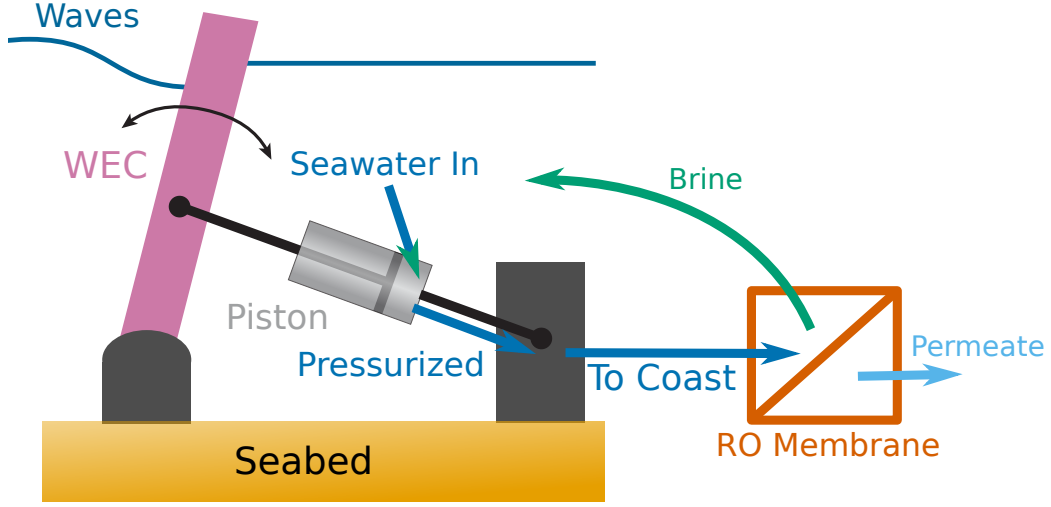


Figure 1: Simple wave-driven desalination system concept sketch.

disciplinary Design Optimization (MDO) provides a systematic framework to address this challenge by simultaneously considering the designs and interactions of multiple subsystems and optimizing overall performance rather than isolated improvements [6]. MDO approaches integrating PTO design with hydrodynamics and/or controls has been demonstrated to significantly improve the design of electricity producing WECs [7, 8, 9, 10]. Michelén Ströfer et al. (2023) see a 22% increase in electrical power output through PTO and control co-design (CCD) [7], while Grasberger et al. (2024) report a 60% improvement in performance using MDO when compared to traditional design methods [10]. These studies emphasize the importance of utilizing holistic models that account for wave dynamics, PTO performance, and control strategies in WEC optimization. These studies show great promise for MDO to improve electricity-generating WEC design, however, these approaches have yet to be applied to wave-driven desalination systems, leaving a critical gap in research.

This paper addresses this gap through the development of a holistic MDO framework for wave-driven desalination system design. A simplified wave-driven desalination schematic is shown in Figure 1, where an oscillating surge wave energy converter (OSWEC) drives a hydraulic cylinder, pressurizing seawater and delivering it to an onshore SWRO desalination plant. This system enables freshwater production without the need for major external energy input. While similar systems have been studied in literature [11, 12, 13, 14], this study has a notable difference in that it lacks an energy recovery unit (ERU).

The concept sketch (Figure 1) depicts a high level overview of the wave-

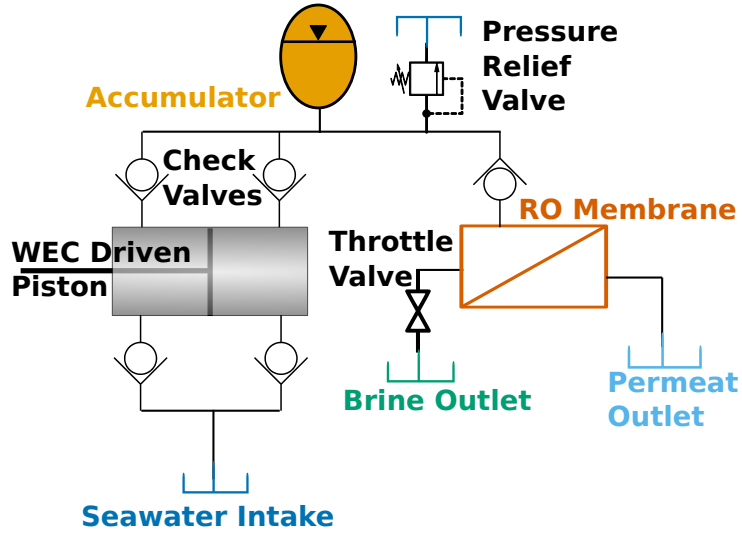


Figure 2: Hydraulic circuit diagram of the wave-driven desalination system.

driven desalination system but does not capture all the hydraulic components included in this design study. A detailed hydraulic circuit diagram is shown in Figure 2, which includes an accumulator to smooth the pressure delivered to the SWRO plant, a pressure relief valve to avoid operating the SWRO plant beyond its rated capacity, a brine side throttle valve set to control the recovery ratio of the SWRO plant, as well as directional valves.

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